



Facilitation

Rationale

The development of an effective nutrient management strategy requires knowledge of mineral nutrition in rice production. Of particular importance are the essential nutrients and their functions in rice growth and development.

The following lessons focus on:

- The essential nutrients and their functions and deficiency symptoms in rice production
- An introduction of the principles of site-specific nutrient management (SSNM) and the steps in determining fertilizer requirements
- Explain the guidelines and procedures for using the leaf colour chart (LCC) to determine the N status of leaves in a standing rice crop

Objectives

Specifically, participants will have a basic understanding of the nutrition requirements of the rice crop and the importance of essential nutrients.

Specific objectives

At the end of this lesson, participants will be able to describe:

- Functions of essential nutrients in rice growth and development, as well as deficiency and toxicity symptoms of essential nutrients;
- Principles of SSNM
- Steps involved in determining fertilizer requirements through the SSNM approach
- Important features of the LCC and the guidelines for using it

Training/learning methodology

Some of the methods used will be:

- Trainer presentation
- Classroom discussion
- Field demonstrations
- Hands-on training

- Required reading

Resource requirements

- White board
- Flip chart
- Marker pens
- Overhead projector
- Computer
- Presentation
- Leaf colour chart

Time requirement: 2 days

Lesson Plan

Content	Method
• Introduction (review what is to come) • Outline key concepts	Power Point
Specific objective 1. Essential nutrients for rice	
The trainer should explain the essential nutrients required -both the macro- and the micro elements.	• Power Point • White board • Flip chart
Specific objective 2. Nitrogen (N) and its importance	
The trainer should explain the importance of N to the crop, focusing on the following: • N-deficiency symptoms • Sources of N • Application of N	• Power Point • White board • Flip chart
Specific objective 3. Phosphorus (P) and its importance	
The trainer should explain the importance of P to the crop, focusing on the following: • P-deficiency symptoms • Sources of P • Application of P	• Power Point • White board • Flip chart

Specific objective 4. Zinc (Zn) and its importance	
The trainer should explain the importance of Zn to the crop, focusing on the following: • Zn-deficiency symptoms • Sources of Zn • Application of Zn	• Power Point • White board • Flip chart
Specific objective 5. Sulfur (S) and its importance	
The trainer should explain the importance of S to the crop, focusing on the following: • S-deficiency symptoms • Sources of S • Application of S	• Power Point • White board • Flip chart
Specific objective 6. Iron (Fe) and its importance	
The trainer should explain the importance of Fe to the crop, focusing on the following: • Fe-deficiency symptoms • Sources of Fe • Application of Fe	• Power Point • White board • Flip chart
Specific objective 7. Calcium (Ca) and its importance	
The trainer should explain the importance of Ca to the crop, focusing on the following: • Ca-deficiency symptoms • Sources of Ca • Application of Ca	• Power Point • White board • Flip chart
Specific objective 8. Magnesium (Mg) and its importance	
The trainer should explain the importance of Mg to the crop, focusing on the following: • Mg-deficiency symptoms • Sources of Mg • Application of Mg	• Power Point • White board • Flip chart
Specific objective 9. Manganese (Mn) and its importance	
The trainer should explain the importance of Mn to the crop, focusing on the following: • Mn-deficiency symptoms • Sources of Mn • Application of Mn	• Power Point • White board • Flip chart

Specific objective 10. Boron (B) and its importance	
The trainer should explain the importance of B to the crop, focusing on the following: • B-deficiency symptoms • Sources & application of B	• Power Point • White board • Flip chart
Specific objective 11. Molybdenum (Mo), chlorine (Cl), and copper (Cu), and their importance	
The trainer should explain the importance of Mo, Cl, and Cu to the crop, focusing on the following: • Cu-, Mo-, and Cl-deficiency symptoms • Sources of Cu, Mo, and Cl • Application of Cu, Mo, and Cl	• Power Point • White board • Flip chart
Specific objective 12. Organic fertilizers and their importance	
The trainer should explain the importance of organic fertilizers to the crop, focusing on the following: • Source • Advantages and disadvantages • Comparison between organic and inorganic fertilizers in terms of nutrients	• Power Point • White board • Flip chart
Specific objective 13. Site-specific nutrient management (SSNM)	
The trainer should explain the importance of SSNM to the crop and should explain the process of using SSNM, including: • How to establish a sustainable yield target • How to effectively use existing nutrients • How to calculate the deficit between crop requirement and availability through indigenous supply and how to apply the right amount of fertilizer	• Power Point • White board • Flip chart

Specific objective 14. Leaf colour chart (LCC)

The trainer should explain the concept behind the LCC and its use.

- Power Point
- White board
- Flip chart
- LCC

Field exercises

1. Use an LCC in the field
2. Look for NPK deficiencies in the field or glasshouse and discuss potential recommendations
3. Examine a range of different fertilizer bags and determine the active ingredients
4. Calculate fertilizer application amounts for different field sizes and application techniques (e.g., machine and broadcast)

Evaluation questions and exercise

1. For the primary macronutrients (N, P, and K), in the plant and the common symptoms associated with a deficiency
2. Name common N and P fertilizer sources
3. How much N, P, and K does the rice plant require to produce 1 ton of grain?
4. What are the factors for the conversion of K into K_2O and P into P_2O_5 , and vice versa?
5. What is the most important micronutrient deficiency in rice?
6. Describe the major symptoms of sulfur deficiency.
7. Under which field conditions can Fe toxicity be a major constraint to rice growth?
8. What are the advantages and disadvantages of organic fertilizers?
9. Explain the basics of SSNM and the necessary steps for applying it.
10. What are the benefits of using the LCC for N management?
11. How much N fertilizer should you apply with an LCC reading of '4'?

Technical content

Introduction

There are 16 essential elements needed for rice production. These may be supplied by the soil or added in the form of chemical or organic

fertilizer. If any of these nutrients are in short supply, the plant will not reach its yield potential. If nutrients are oversupplied or applied at an incorrect time, these can have detrimental effects on plant growth and may cause problems such as lodging, increase in pests and diseases, or environmental pollution.

Essential nutrients

The 16 essential nutrients can be grouped into macro- and microelements. The macroelements are carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, calcium, magnesium, and sulfur. All of these are needed by the plants in large amounts, often more than 0.1% of the plant's dry weight.

The microelements are iron, manganese, copper, zinc, molybdenum, boron, and chlorine. These are needed by the plants in lesser quantities (trace amounts). Silicon is also classified as a 'beneficial' nutrient for rice, but its physiological functions are not yet clear.

Carbon, hydrogen, and oxygen

Carbon, hydrogen, and oxygen, which are common to all organic matter, are obtained by the plant from carbon dioxide (CO_2) and water (H_2O) through the photosynthesis.

The other essential elements are supplied from the soil and are either naturally occurring or added in the form of fertilizer.

Nitrogen (N)

Nitrogen is an essential plant nutrient, being a component of amino acids, nucleic acids, nucleotides, chlorophyll, enzymes, and hormones. Nitrogen promotes rapid plant growth and improves grain yield and grain quality through higher tillering, leaf area development, grain formation, grain filling, and protein synthesis. Nitrogen is highly mobile within the plant and soil and is the most limiting element in almost all soils.

Rice plants require a large amount of nitrogen at the early and mid-tillering stages to maximize the number of panicles. Nitrogen absorbed at the panicle initiation stage increases the spikelet

number per panicle, and nitrogen absorbed during the ripening stage enhances the grain-filling process.

Nitrogen deficiency:

The symptoms of nitrogen deficiency include:

- Stunted yellowish plants with limited number of tillers
- Narrow, short, erect leaves that become yellowish green as they age
- Old leaves and, sometimes, all leaves, become light green and chlorotic at the tip, with leaves dying under severe nitrogen stress
- Early maturity or shortened growth duration.

Sources of N:

At optimum levels of nutrition, the rice crop takes up around 16 kg N/t of grain yield - 10 kg N in the grain and 6 kg N in the straw. Chemical or inorganic nitrogen fertilizer is the most common source of N and is normally the most cost-effective. Organic N available in farmyard manure and compost can be effective in partially meeting some of the N requirement. The N available from most organic sources is between 1 and 2%. The most commonly used inorganic fertilizers, which contain N, are ammonium sulfate (21% N, 24% S), urea (46% N), and ammonium phosphate (18% N 44-46% P_2O_5).

Application of N:

Nitrogen is best applied in three- or four-split applications. Depending on the establishment technique and local environment, a basal dose of at least 10-20% N should be applied at planting or early establishment. Another 30-40% should be applied 30 days after establishment, and the remainder at panicle initiation or just before flowering.

A good rule of thumb is to apply 40-50 kg N/ha as inorganic fertilizer for every 1 ton of additional grain yield over the yield without N. That means for most soils, growing a 5 ton/ha crop will require between 80 and 90 kg N/ha.

Nitrogen deficiency symptoms:



- Leaves are yellowish green in the 0 N omission plot, since fertilizer is not applied.
- Leaves of N-deficient plants are light green, narrow, and smaller.
- Tillering is reduced when N is deficient.
- Tillering is greater when N fertilizer has been applied.

Phosphorus (P)

Phosphorus is used in the supply, transfer, and storage of energy for all biochemical processes in the plant. It is an essential constituent of adenosine triphosphate, nucleotides, nucleic acids, and phospholipids.

Its specific functions include a major role in the formation of plant hormones and maintenance of membrane integrity, root development, and promoting rapid growth of the plant, especially during active tillering and early flowering. It also hastens ripening and grain development. P is often deficient in sandy soils with low organic matter.

Phosphorus deficiency:

The symptoms include the following:

- Stunted dark green plants with limited number of tillers

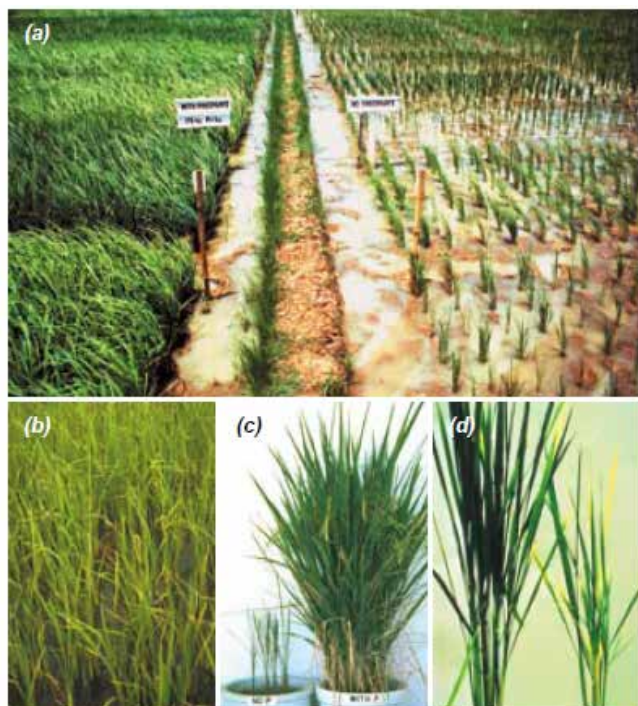
- Narrow, dark green, stiff, and erect leaves with young leaves remaining healthier than older leaves, which turn brown and die
- Reddish or purplish colour that may develop on leaves (tend to produce anthocyanin pigment)
- Leaves that appear pale green when P and N deficiencies occur simultaneously

Sources of P:

At optimum plant nutrition, the rice crop takes up around 2.8 kg P/t (6.4 kg P_2O_5 /t) of grain yield - 1.9 kg P in the grain and 0.9 kg P in the straw. Phosphorus fertilizer sources and each fertilizer's P_2O_5 equivalent are given in the following table.

		Fertilizer P ₂ O ₅ (kg/ha)				
P fertilizer	% P ₂ O ₅	15	20	30	40	60
		Amount of fertilizer required (kg/ha)				
Single superphosphate	16–18	88	117	176	234	352
Triple superphosphate	44–46	33	44	66	88	132
Diammonium phosphate (DAP)*	44–46	33	44	66	88	**

Phosphorus deficiency symptoms:



- Tillering is reduced where P is deficient
- Even under less pronounced P deficiency, stems are thin and spindly and plant development is retarded
- d. Plants are stunted, small, and erect when compared to normal plants

Application of phosphorus:

Recommendations for P are based on crop yield and soil type. A 5-ton crop will need 15-20 kg P or 30-50 kg P_2O_5 , depending on whether the straw is removed. All P fertilizer should be incorporated before the last soil puddling before transplanting or as topdressing within 10-15 days after direct seeding when permanent water is applied.

Potassium (K)

Potassium plays an essential function in osmoregulation, enzyme activation, regulation of cellular pH, cellular cation-anion balance, regulation of transpiration by stomata, and transport of photosynthetic products. Potassium is an essential plant nutrient that improves root growth and plant vigor, helps prevent lodging, and enhances crop resistance to pests and diseases. Potassium is mobile in the plant and quite mobile in the soil.

Potassium is often the most limiting nutrient after N in high-yielding rice systems. Potassium deficiency occurs in intensively cropped areas with high levels of N and P application and is often deficient in coarse-textured sandy soils. Potassium fertilizer needs to be applied in most irrigated rice fields.

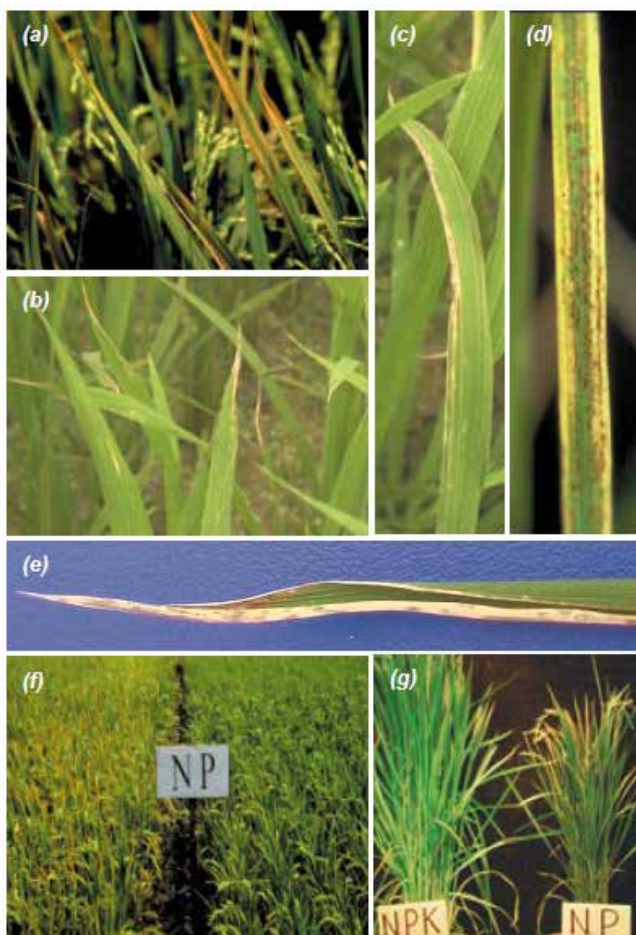
Potassium deficiency:

The symptoms include:

- Dark green plants with yellowish brown leaf margins or dark brown necrotic spots first appearing on the tips of the older leaves
- Stunted plants, tillering slightly reduced, weak straw, droopy leaves
- Wilting when there is excessive imbalance with N (low K-N ratio) in the plant



Potassium-deficiency symptoms:



- a, b, c. Leaf tips and margins become yellowish brown and dry up under K deficiency
- d. Plants are more susceptible to pests and diseases, and secondary infections are common
- e. Leaf rolling may occur
- f. Hybrid rice produces more biomass and, therefore, has a greater K requirement than inbred rice; so K-deficiency symptoms may occur earlier in hybrid (left) than inbred rice (right)
- g. Plant growth is restricted in the absence of K

Application of K:

The rice crop takes up around 15.8 kg K/t (19 kg K_2O) of grain and, of this, 1.8 kg K is in the grain and 14 kg K is in the straw. If the required K rates are less than 30 kg K/ ha, all K should be incorporated before the last soil puddling before transplanting or topdressed within 10-15 days after direct seeding. If rates higher than 30 kg K/ ha are needed, 50% should be applied at basal and 50% at early panicle initiation. K should be applied in two doses if the soil is sandy and prone to leaching. The use of K at heading to early flowering increases resistance to lodging and diseases in dense canopies.

Sources of potassium:

The most common K fertilizer is muriate of potash (MOP), which contains about 50% K or 60% K_2O . Rice straw is rich in K (14.0 kg K or 16.8 kg K_2O /t of straw).

Zinc (Zn)

Zinc plays an important role in auxin metabolism, N metabolism, cytochrome and nucleotide synthesis, chlorophyll production, and enzyme activation. It is also an essential component in the cellular membrane and is present in some essential enzymes. Zn deficiency affects plant colour and turgor. Zn is moderately mobile in the plant but relatively immobile in the soil.

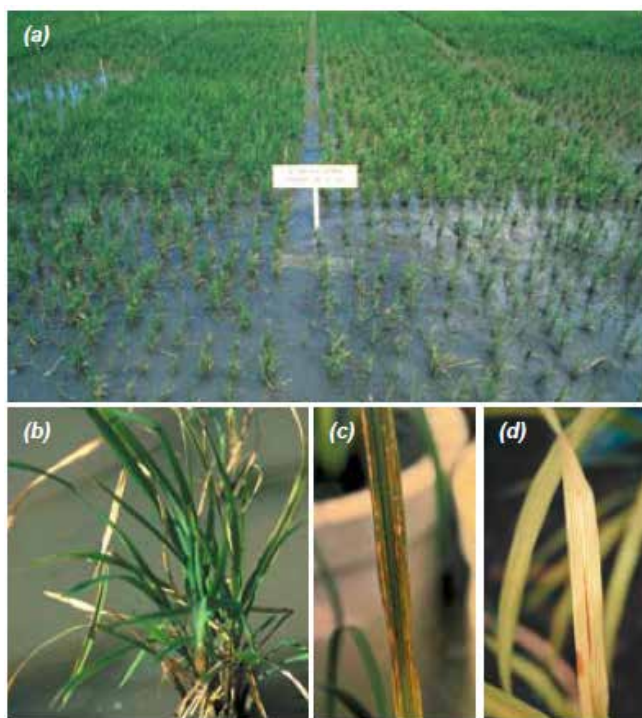
Zinc deficiency:

The symptoms include:

- Dusty brown spots on the upper leaves of stunted plants appearing 2-4 weeks after transplanting
- Midribs near the leaf base of younger leaves become chlorotic; leaves lose turgor and turn brown as blotches appear on the lower leaves
- Presence of uneven plant growth and patches of poorly established hills in the field at 2-4 weeks after transplanting
- Decreased tillering, which may stop completely, and longer time to maturity or lengthened growth duration of the plant under severe zinc deficiency
- Increased spikelet sterility



Zinc-deficiency symptoms:



- a. Uneven field with stunted plant growth (foreground)
b. Tillering is reduced, leaves are droopy and dry
c, d. Appearance of dusty brown spots and streaks

Application of Zn:

If Zn deficiency symptoms are observed in the field, apply 10-25 kg ZnSO_4 into water or 20-40 kg ZnSO_4 on the soil surface. Roots of rice seedlings may be dipped in a 2-4% Zn suspension (20-40 g ZnO /liter water) before planting. Rice plants can recover from Zn deficiency if the field is drained as aerobic soil conditions increase the availability of Zn.

The crop only requires around 0.05 kg Zn/ha per ton of grain yield, but much more Zn fertilizer must be applied because Zn is not readily available. Zn can be applied before puddling in the main field, or to the nursery beds 7-8 days before pulling the seedlings. The effect of a Zn application can last two to five cropping seasons on all soils, except in alkaline soils. In alkaline soils, Zn may need to be applied to each crop.

Sources of zinc:

The commonly used Zn fertilizers are soluble zinc sulfate (23-36% Zn), soluble zinc chloride (48-50% Zn), and insoluble zinc oxide (60-80% Zn).

Sulfur (S)

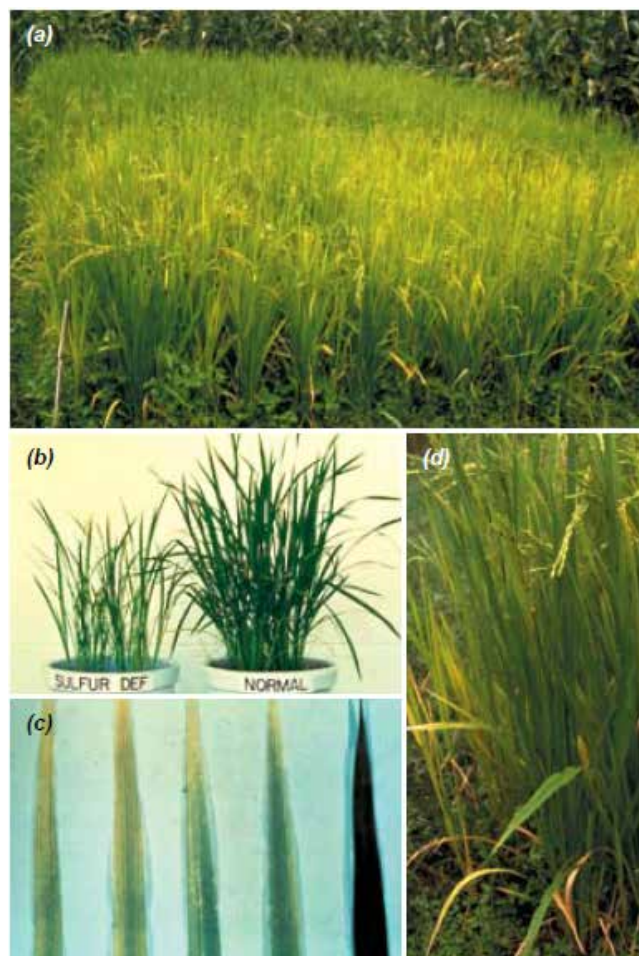
Sulfur is an essential constituent of amino acids (cystine, cysteine, and methionine) involved in chlorophyll production and protein synthesis and acts as an essential component of plant structures. It is also an important component of thiamine and biotine, which are both involved in carbohydrate metabolism. It is also involved in the functioning of many plant enzymes, enzyme activators, and oxidation-reduction reactions. Sulfur is quite mobile in the plant, but only partially mobile in the soil.

Sulfur deficiency:

The symptoms include:

- There is yellowing of the whole plant with chlorosis being more pronounced in young leaves possibly with necrosis of the tips; this is in contrast with N deficiency, wherein older leaves are affected first

Sulfur-deficiency symptoms:



- a, b. The leaf canopy appears pale yellow because of yellowing of the youngest leaves, and plant height and tillering are reduced
c, d. Chlorosis is more pronounced in young leaves, where the leaf tips may become more necrotic

- In comparison with N-deficient plants, no necrosis in lower leaves is seen and leaves are generally paler in colour
- Plant height and tiller number get reduced, with fewer and shorter panicles, and reduced numbers of spikelets per panicle at maturity
- Plant development and maturity get delayed by 1 - 2 weeks

Application of S:

Sulfur deficiency is not very common, but S may be needed in leached sandy soils, which have low organic matter. When needed, 10 kg S/ha should be applied for moderate S deficiency and 20 - 40 kg S/ha for severe deficiency. The crop requires around 2 kg S/ha for both straw and grain per ton of grain yield. S fertilizer should be applied before the last puddling along with P and K. The effect of an S application may last for two cropping seasons.

Sources of S:

The commonly used S fertilizers are ammonium sulfate (24% S), single superphosphate (12% S), and gypsum (17% S).

Iron (Fe)

Iron is an essential plant nutrient required for electron transport in photosynthesis. It is an important electron acceptor in redox reactions and an activator for several enzymes. Iron deficiency may inhibit K absorption. Iron is not mobile in the plant or soil.

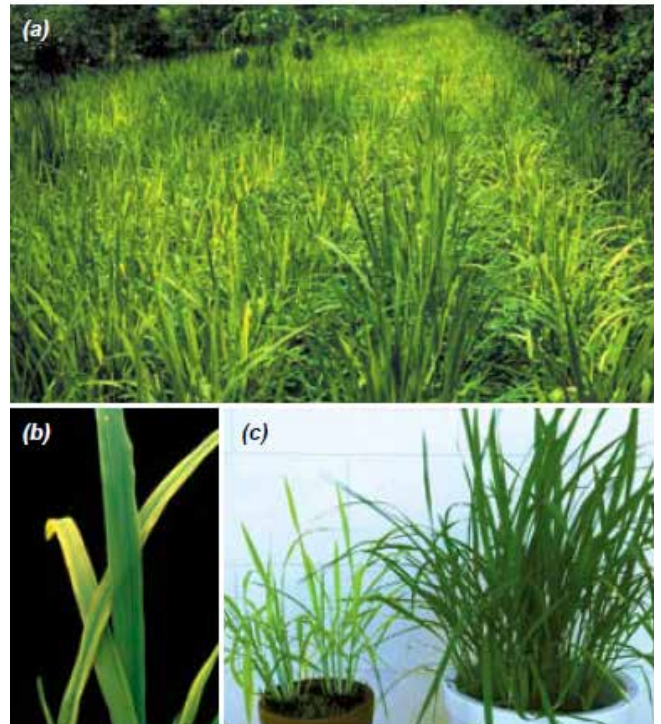
Iron deficiency:

The symptoms include:

- Interveinal yellowing and chlorotic leaves that turn whitish
- Chlorotic and, ultimately, dead plants (in cases of severe deficiency)



Iron-deficiency symptoms:



- Fe deficiency is mainly a problem on upland soils
- Interveinal yellowing of emerging leaves
- Plants are stunted and have narrow leaves (left) if Fe deficiency is severe

Application of Fe:

Iron deficiency is not very common in flooded or slightly acidic lowland soils, but sandy, calcareous, and alkaline soils may suffer from Fe deficiency. Dry-seeded rice crops often suffer from Fe deficiency when grown on lighter soils and the deficiency is worse in low-rainfall seasons. The symptoms generally appear during the early vegetative stage. Iron deficiency is the most difficult and costly nutrient deficiency problem to correct. Soil application requires 100-300 kg/ha. Foliar applications of 2-3% ferrous sulfate solution or iron chelates (200 liter/ha) can be used two to three times at 2-week intervals starting at tillering. The crop requires around 0.5 kg Fe/ha (both straw and grain) per ton of grain yield but, once applied, Fe is not freely available to the plant.

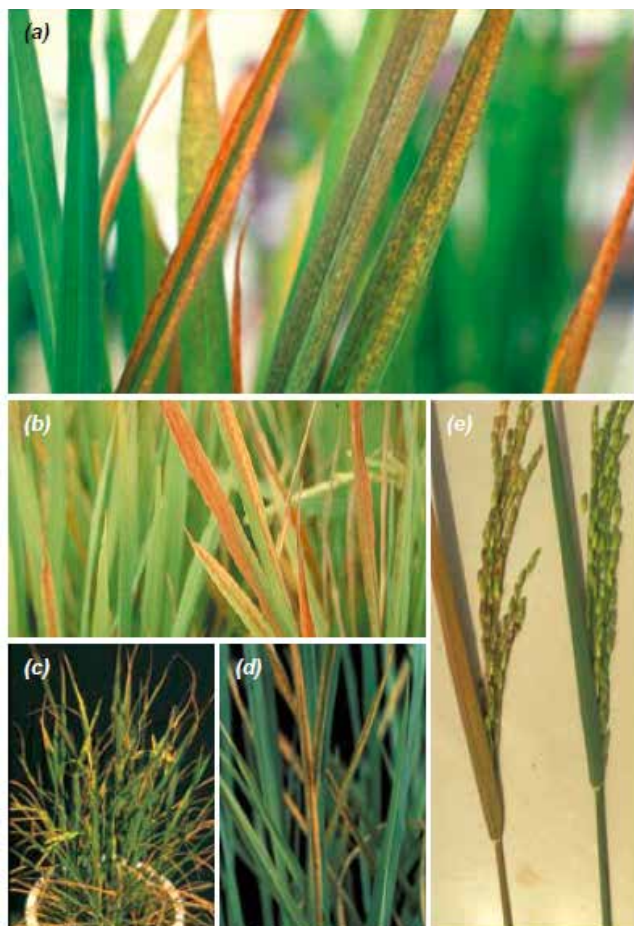
Sources of Fe:

The commonly used iron fertilizers are soluble ferrous sulfate (20-33% Fe), ferrous ammonium sulfate (14% Fe), and iron chelates (5-14% Fe).

Iron toxicity:

Iron toxicity is often found in areas where the soil is permanently waterlogged or saturated. Allowing the soil to dry by improved drainage or imposing alternate wetting and drying techniques will help alleviate the problem.

Iron-toxicity symptoms:



- a. Tiny brown spots develop on the leaf tip and spread toward the leaf base
- b. Leaves turn orange-brown and die
- c. Symptoms first appear in older leaves
- a, d. Under severe Fe toxicity, the whole leaf surface is affected
- e. Leaf bronzing (left) compared to healthy plant (right)

Iron toxicity symptoms include:

- Tiny brown spots appear on lower leaves, starting from the tips and spreading toward the bases. Spots generally combine on interveins
- Leaves are narrow and remain green, but in severe cases become purplish brown (“bronzing”)
- The root system is coarse, sparse, and damaged with dark brown to black coating on the root surface and many dead roots

Calcium (Ca)

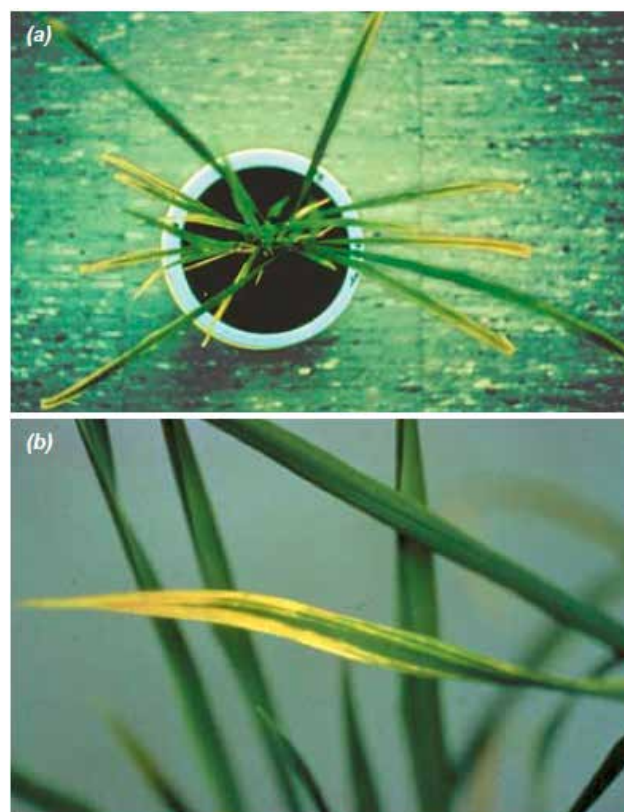
Calcium acts as an important constituent of calcium pectate, which is important for the maintenance of biomembranes and cell wall integrity. It also acts as an enzyme activator and is required for osmoregulation and maintenance of the cation-anion balance in cells. It helps promote normal root growth and development.

Calcium deficiency:

Calcium deficiency causes little change in the general appearance of the plant, except in cases of acute deficiency when:

- The tip of the upper growing leaves becomes white, rolled, and curled
- Necrotic tissue develops along the lateral margins of leaves, which eventually turn brown and die
- In extreme cases, the plant becomes stunted and the growing point dies
- Symptoms resemble those of boron deficiency, and requires tissue analysis to identify the cause of symptoms

Calcium-deficiency symptoms:



- a, b. Symptoms occur only under severe Ca deficiency, when the tips of the youngest leaves may become chlorotic-white

Magnesium (Mg)

Magnesium acts as a constituent of chlorophyll involved in CO₂ assimilation and protein synthesis and activates several essential enzymes. It acts in a manner similar to calcium in the regulation of cellular pH and maintenance of the cation-anion balance in cells.

Magnesium deficiency:

The symptoms include:

- Pale-coloured plants, with interveinal chlorosis first appearing on older leaves and later on the younger leaves as deficiency becomes more severe
- In contrast with K deficiency (where green and yellow stripes run parallel to the leaf), green colouring appears as strings of beads; in severe cases, chlorosis progresses to yellowing and finally, necrosis in older leaves
- Reduced number of spikelets, grain weight, and grain quality (% of milled rice, protein, and starch content), height and tiller number not affected much at low to moderate deficiency ranges

Magnesium-deficiency symptoms:



- Orange-yellow interveinal chlorosis usually appears on older leaves.
- Chlorosis may also occur on the flag leaf.
- Mg deficiency may also be induced by large applications of K fertilizer on soils with low Mg status.

Manganese (Mn)

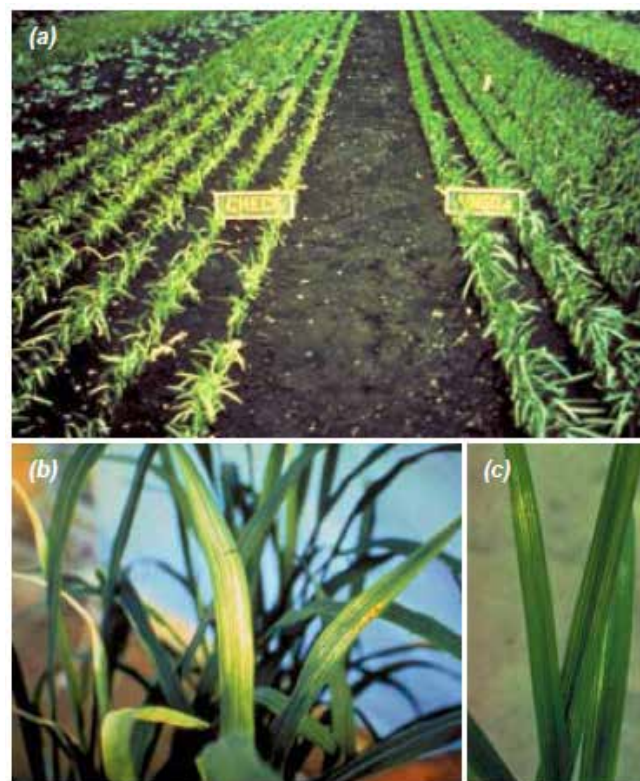
Manganese plays an important role in O₂ evolution in photosynthesis, and in the oxidation-reduction processes that involve electron transport systems. It is also involved in activating several enzymes such as oxidase, peroxidase, dehydrogenase, decarboxylase, and kinase. Manganese enables the formation and stability of chloroplast, protein synthesis, nitrate reduction, and tricarboxylic cycle, and helps reduce Fe toxicity.

Manganese deficiency:

The symptoms include:

- Interveinal chlorosis, which begins at the tip of the younger leaves
- Chlorotic streaks that become dark brown and necrotic and spread downward from the tip to the base of the leaves
- Newly emerging leaves, which are short, narrow, and light green
- Stunting (though tillering is not affected)

Manganese-deficiency symptoms:



- Deficiency is mainly a problem in rice grown in upland and organic soils with low Mn status
- Leaves are affected by interveinal chlorosis that appears at the tip of the younger leaf

Manganese toxicity:

The symptoms include:

- Stunted plants with limited tillers
- Brown spots on the veins of the leaf blade and leaf sheath, particularly in lower leaves

Boron (B)

Boron plays a primary role in cell wall biosynthesis and structure, as well as plasma membrane integrity. It is also involved in regulating or catalyzing physiological functions, such as carbohydrate metabolism, nitrogen metabolism, nutrient uptake, sugar transport, lignification, nucleotide synthesis, and respiration.

Boron deficiency:

The symptoms include:

- Reduced plant height
- White, rolled tips of emerging leaves as in cases of Ca deficiency
- Dead growing points (in severe cases), although new tillers continue to be produced
- Reduced or arrested panicle formation

Boron toxicity:.

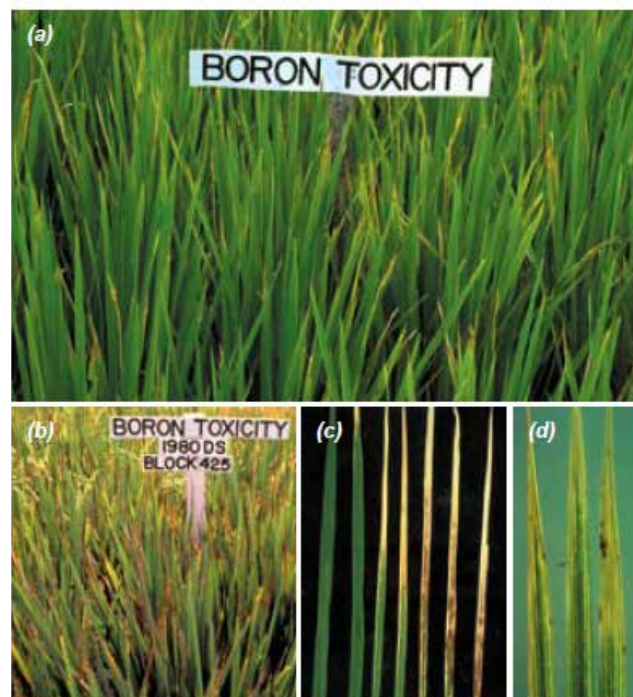
The symptoms include:

Chlorosis at the tips of older leaves, especially along the margins. This is followed by the appearance of large, dark brown elliptical spots in the affected parts that ultimately turn brown and dry up.

Symptom development occurs as follows: (A) yellowish white discolouration of tips of older leaves appearing 6 weeks after transplanting; (B) as toxicity progresses, yellowing of tips and leaf margins; (C) depending on severity and variety, elliptical dark brown blotches appearing in discoloured areas; and (D) entire leaf blades turning light brown and withering.



Boron-toxicity symptoms:



a. Brownish leaf tips is a typical characteristic of B toxicity, appearing first as marginal chlorosis on the tips of older leaves

b, c, d. Two to four weeks later, brown elliptical spots develop on the discoloured areas

Molybdenum (Mo)

Molybdenum's function in rice plants is limited to the reduction of nitrate to nitrite. No deficiency symptoms have been identified.

Copper (Cu)

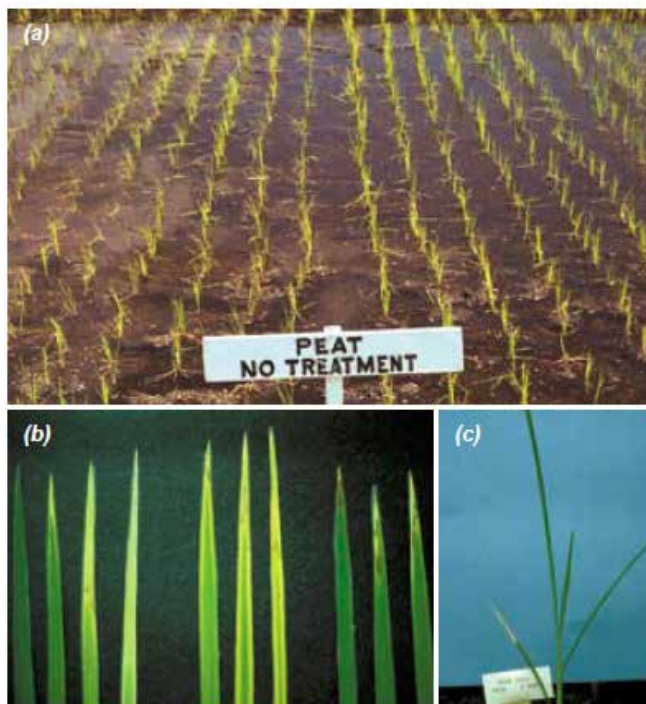
Copper acts as a component of metalloenzymes, regulating some enzymatic actions, and catalyzing oxidation reactions. It also plays a role in nitrogen, protein and hormone metabolism, photosynthesis, respiration, and pollen formation and fertilization.

Copper deficiency:

The symptoms include:

- Bluish green leaves, which become chlorotic near the tips
- Development of chlorosis downward along both sides of the midrib, followed by dark brown necrosis of the leaf tips
- New leaves failing to unroll and maintaining a needle-like appearance of the entire leaf or occasionally of half the leaf, with the basal portion developing normally.

Copper-deficiency symptoms:



a. Deficiency mainly occurs in organic soils

b. Chlorotic streaks and dark brown necrotic lesions may develop on the tips of younger leaves

c. New leaves may have needle-like appearance

Chlorine (Cl)

Chlorine is essential for photosynthesis. Deficiency symptoms have not been described in rice.

Organic fertilizers

Organic fertilizers are materials that come from plant or animal waste or their by-products. These include:

- Cattle or poultry manure
- Composted rice straw or other crop residues
- Sewage sludge
- Oil cakes
- Green manure (plant material like sesbania can be incorporated into the soil while green)
- Biofertilizers including azolla, blue green algae, rhizobium, azotobacter, azospirillum, etc. can be used during different crop growth stages

The advantages of using animal manure and other organic material are they:

- Are low cost
- Contribute to improving soil structure and fertility

- Provide micronutrients not normally supplied through inorganic fertilizers
- Enhance microbial growth and nutrient turnover in soil
- Improve the water-holding capacity of soil

The following are the limitations of using organic fertilizers.

- Organic material can be bulky, with high handling and transport costs
- Organic manure is sometimes more expensive than inorganic fertilizers, per unit of nutrient
- Organic fertilizers are not always available
- Organic fertilizers must be applied at the beginning of the crop, so early applications may not meet later crop demand for nutrients
- Organic material can smell bad



Organic fertilizers are normally applied uniformly across the field, 2 or more weeks before being incorporated into the soil during land preparation. Sometimes, rice straw is directly composted in the field. Organic fertilizers are commonly used in seedling nurseries, where they improve nutrient status and make seedlings easier to pull.

** The carbon nitrogen (CN) ratio will be the key indicator of the authenticity of organic input in the farmer's field.*

Below is a comparison between different fertilizer sources that apply an equivalent of 100 kg N/ha:

Urea (46% N)	Poultry manure (2% N)	Cattle manure (1% N)	Compost (0.5% N)
217 kg	5,000 kg or 5 t	10,000 kg or 10 t	20,000 kg or 20 t

In some instances, inorganic N fertilizers are added when composting plant material to improve the carbon-nitrogen ratio aiding nutrient release.

Site-specific nutrient management

Site-specific nutrient management (SSNM) is a plant-based approach that helps farmers to optimize the use of N, P, and K in their rice fields. The growth and nutrient needs of each rice crop will vary not only within and between fields but between seasons and years. With SSNM, nutrients are supplied as and when required by the crop

Steps in applying SSNM

Step 1. Establish an attainable yield target

Rice yields are location- and season-specific, depending on climate, variety, and crop management. The yield target for a given location and season is the estimated yield attainable with good crop management when N, P, and K are not limiting. The amount of nutrients taken up by a rice crop is directly related to yield. It is the yield target that indicates the total amount of nutrients that should be applied to the crop.

Step 2. Effectively use existing nutrients

SSNM promotes the optimal use of existing (indigenous) nutrients, which come from the soil, organic amendments, crop residue, manure, and irrigation water. The uptake of a nutrient from indigenous sources can be estimated by the grain yield for the crop without applying N, P, or K. This can be determined by leaving a strip in the field where a respective nutrient source is not applied (N, P, or K omission plots).

Step 3. Apply fertilizer to fill the deficit between crop need and indigenous supply

N, P, and K fertilizers are applied to supplement the nutrients from indigenous sources to achieve the yield target. The quantity of fertilizer needed is determined by subtracting the yield of the omission plots from the target yield. As a baseline, each ton of grain will need approximately 20 kg N, 5 kg P, and 5 kg K, if straw is not removed.

In SSNM, fertilizers are applied using the following principles to achieve high yield and high efficiency of plant use:

1. Apply 10-20% of N fertilizer to young rice within 14 days after transplanting (DAT) or 21 days after sowing (DAS), when the need of the crop for supplemental N is small.
2. Apply 30-40% N fertilizer at 30-35 days after establishment based on the crop's need for N, as determined by leaf N status. Apply the remaining 30-40% of N at panicle initiation. If the crop is showing signs of N deficiency during the tillering stage, it may pay to split the final N application. The leaf colour chart (LCC) is a tool that can be used for assessing leaf N status and the crop's need for N. (Important: Do not apply N into dry fields.)
3. Apply all P fertilizer near transplanting or sowing.
4. Apply K fertilizer twice-50% close to the transplanting or sowing stage, and 50% at early panicle initiation. When K fertilizer rates are relatively low (for example, ≤ 30 kg K_2O/ha), it can all be applied close to the time of transplanting or sowing.
5. SSNM must be adjusted to local needs taking into consideration cropping history, seasonal effects, irrigation strategies, and expected weather patterns.

Rice Crop Manager (RCM) for Odisha

The Rice Crop Manager aims to sustain the productivity of rice-based cropping systems and increase the net income of farmers in Odisha, India. It achieves this by providing irrigated and rainfed rice farmers with crop and nutrient management guidelines customized to the needs of the individual farmer. The Rice Crop Manager (<http://webapps.irri.org/in/od/rcm/>) uses a farmer's answers to questions on his or her rice farming practices to automatically generate rice crop management guidelines aimed at increasing the farmer's net income. The Rice Crop Manager was adapted, evaluated, and verified for rice cultivation in Odisha on a collaborative basis by IRRI, Odisha University of Agriculture and Technology (OUAT) and the National Rice Research Institute (NRRI). The adoption and scaling of this SSNM tool will enhance nutrient

use efficiency, improve productivity and reduce the environmental footprint.

Leaf colour chart

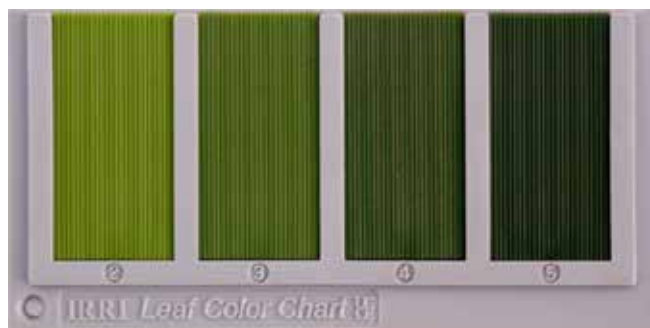
The leaf colour chart (LCC) is also a simple, easy-to-use, and inexpensive tool that can help farmers monitor the relative greenness of a rice leaf as an indicator of plant N status.

It helps farmers optimize N use, regardless of the source of N applied-organic, such as manure and crop residue, or inorganic, such as urea or NPK compound fertilizers.

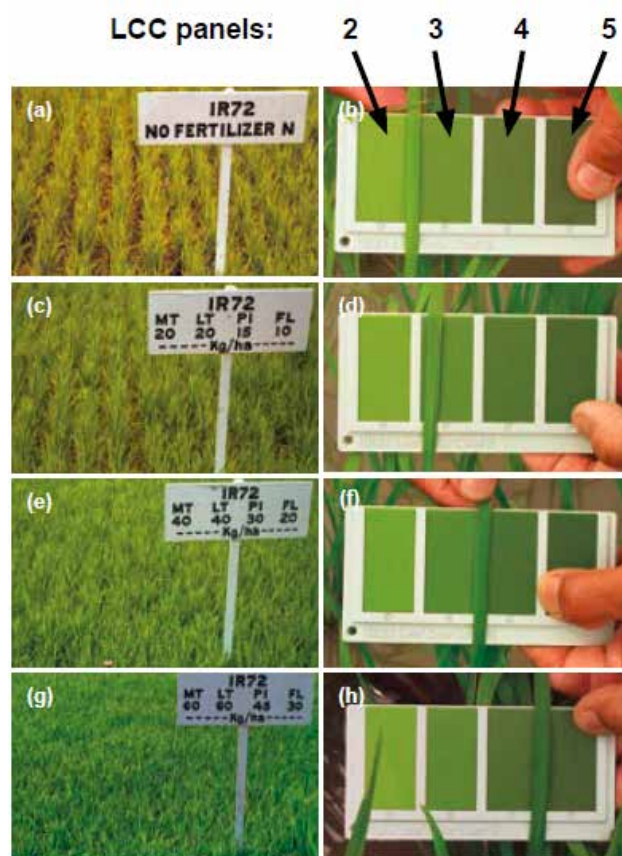
For efficient use of this input, farmers should apply N fertilizer several times during the growing season to ensure that N supply matches crop need at critical growth stages. In this way, N fertilizer losses are minimized.

Using the LCC

- Randomly select at least 10 disease-free rice plants or hills in a field with uniform plant population
- Select the topmost fully expanded leaf from each hill or plant. Place the middle part of the leaf on a chart and compare the leaf colour with the colour panels of the LCC. Do not detach or destroy the leaf.
- Measure the leaf colour under the shade of your body because direct sunlight affects leaf colour readings. If possible, the same person should take LCC readings at the same time of day, every time.
- Determine the average LCC reading for the selected leaves



The photos here illustrate the use of the standardized LCC to assess leaf N status and adjust N applications to rice:



In photos (a) and (b), the plants were cultivated with low N fertilizer rates. The corresponding low LCC reading of 3 indicates N deficiency.

At higher N fertilizer rates in photos (c) and (d), the plants look well developed and the canopy is closed. The LCC reading is between panels 3 and 4, which is the critical range for most transplanted rice.

In photos (e) and (f), plants with a high N rate are dark green. Leaf colour is darker than LCC panel no. 4, indicating a surplus of N fertilizer. This crop does not require N fertilizer.